

Difference-in-Differences

Samuel Rowe adapted from Cunningham (2022)

2026-04-15

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bookdown::render_book()

Overview

Topics:

1. Difference-in-Differences
2. Placebo Tests: False Treatment
3. *xtdidregress* command
4. Post-estimation tests
5. Placebo Tests: False Timing

Chapter 1

Difference-in-Differences

1.1 Background

Cunningham and Cornwell (2013) use a difference-in-difference design to test the abortion hypothesis on long-term gonorrhea incidences for 15-19 year olds. Cunningham (2021) makes a good point with good theories provide very specific falsifiable hypotheses. More specific the hypothesis, the more compelling the theory if evidence supports the theory.

The abortion hypothesis from Gruber, Levine, and Staiger (1999) makes very specific predictions about reproductive health and poverty outcomes. If there are far-reaching effects of abortion legalization in the early 1970s, then some of those effects will show up later. Levitt (2004) finds controversial evidence that abortion legalization caused a 10% decline in crime between 1991 and 2001. Cunningham and Cornwell (2013) use a difference-in-difference design as opposed to Donohue and Levitt (2001) that use lagged ratio values at the state level.

The authors' focus on long-term gonorrhea cases is due to the correlation between single-parent status and risky behaviors. The authors focus on 15-19 year olds, since they would have been treated in the 5 early adopter states before complete legalization.

Five states legalized abortion and then all states are exposed to legalized abortion. There is a three-year lag between the five states and *Roe v Wade* decision, which should lead to a nonlinear parabolic treatment effect.

There should be increasingly negative effect of abortion legalization on the outcome between the treatment states and control states, and then the difference should disappear after the other states are impacted by *Roe v. Wade*.

Our repeal states are Alaska (02), California (06), Hawaii (15), New York (36), Washington (53).

1.2 Estimator

We will use the difference-in-differences estimator. Cunningham and Cornwell (2013) model is the following:

$$Y_{st} = \alpha + \beta_1 \text{RepealST}_s + \beta_2 \text{Year}_t + \delta(\text{RepealST} * \text{Year})_{st} + \psi X_{st} + \gamma ST_s + \varepsilon_{st}$$

Where

1. Y_{st} is outcome of interest: new gonorrhea cases for 15-19 year olds per 100,000
2. δ is our parameter of interest.
3. RepealST_s is a binary for treatment state
4. Year_t is a time binary for each year
5. $\text{RepealST} * \text{Year}_{st}$ is our interaction term
6. X_{st} are a set of covariates, such as alcholo consumption per capita, real income per capita, etc.
7. ST_s are state fixed effects

Our parameter of interest is δ , which is our difference-in-differences estimator.

1.3 Parameter Estimates

We will estimate the *ATE* with the difference-in-differences estimator. We will use the `##` operator for the interaction between treatment state and year. We will use analytical weights to account for population. We will cluster our standard errors by FIPS code (state).

```
use "/Users/Sam/Desktop/Econ 672/Data/abortion.dta", clear
reg ln r i.repeal##i.year i.fip acc ir pi alcohol crack poverty income ur if bf15==1 [a
```

```
(sum of wgt is 43,100,087)
note: 53.fip omitted because of collinearity.
```

Linear regression	Number of obs	=	736
	F(26, 50)	=	.
	Prob > F	=	.
	R-squared	=	0.8566
	Root MSE	=	.24363

(Std. err. adjusted for 51 clusters in fip)

lnr	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.repeal	-.9638237	.3925204	-2.46	0.018	-1.752224	-.1754232
year						
1986	-.0208274	.0576675	-0.36	0.719	-.136656	.0950013
1987	-.2864665	.1109824	-2.58	0.013	-.5093812	-.0635518
1988	-.3494648	.1309365	-2.67	0.010	-.6124586	-.0864711
1989	-.40722	.1732298	-2.35	0.023	-.7551622	-.0592778
1990	-.4864117	.2166425	-2.25	0.029	-.9215509	-.0512725
1991	-.5353034	.2230406	-2.40	0.020	-.9832937	-.0873131
1992	-.7866236	.2685517	-2.93	0.005	-1.326025	-.2472217
1993	-.9767166	.2867949	-3.41	0.001	-1.552761	-.4006721
1994	-1.064367	.3225681	-3.30	0.002	-1.712264	-.4164694
1995	-1.356245	.3752309	-3.61	0.001	-2.109918	-.6025717
1996	-1.520062	.4142987	-3.67	0.001	-2.352206	-.6879188
1997	-1.571846	.467604	-3.36	0.001	-2.511056	-.6326358
1998	-1.545129	.5354385	-2.89	0.006	-2.620589	-.4696694
1999	-1.587414	.5783181	-2.74	0.008	-2.749	-.4258275
2000	-1.672572	.6572878	-2.54	0.014	-2.992773	-.3523703
repeal#year						
1 1986	-.2590538	.06331	-4.09	0.000	-.3862157	-.1318919
1 1987	-.341738	.1683369	-2.03	0.048	-.6798526	-.0036234
1 1988	-.6105532	.2026137	-3.01	0.004	-1.017515	-.2035916
1 1989	-.7850003	.2372121	-3.31	0.002	-1.261455	-.3085458
1 1990	-.6316315	.175649	-3.60	0.001	-.9844328	-.2788301
1 1991	-.5526026	.1393595	-3.97	0.000	-.8325144	-.2726909
1 1992	-.4424166	.1603875	-2.76	0.008	-.7645643	-.1202689
1 1993	-.3060626	.1807604	-1.69	0.097	-.6691306	.0570054
1 1994	-.1178897	.2066789	-0.57	0.571	-.5330165	.2972371
1 1995	.0210291	.2226034	0.09	0.925	-.426083	.4681413
1 1996	-.1235341	.2065672	-0.60	0.553	-.5384365	.2913682
1 1997	.0208529	.2641661	0.08	0.937	-.5097404	.5514461
1 1998	-.0360524	.3473215	-0.10	0.918	-.7336682	.6615635
1 1999	.0147091	.3574973	0.04	0.967	-.7033454	.7327637
1 2000	.0414508	.3873416	0.11	0.915	-.7365477	.8194494
fip						
2	-.8718965	.2991619	-2.91	0.005	-1.472781	-.2710121
4	-.9589386	.5238254	-1.83	0.073	-2.011073	.0931956
5	.086031	.0651941	1.32	0.193	-.0449151	.2169772
6	-.2975866	.2226081	-1.34	0.187	-.7447082	.149535
8	-.824605	.418118	-1.97	0.054	-1.66442	.0152097

9		-1.680892	.735009	-2.29	0.026	-3.157201	-.2045828
10		-.7457544	.4837752	-1.54	0.129	-1.717445	.2259366
11		-2.112355	1.245639	-1.70	0.096	-4.614294	.3895843
12		-1.094748	.5131535	-2.13	0.038	-2.125447	-.0640485
13		-.7215762	.2461082	-2.93	0.005	-1.215899	-.2272534
15		-.785426	.1104538	-7.11	0.000	-1.007279	-.563573
16		-1.793719	.199309	-9.00	0.000	-2.194043	-1.393395
17		-.7228205	.4586163	-1.58	0.121	-1.643979	.1983376
18		-.1431537	.1870356	-0.77	0.448	-.5188258	.2325184
19		-.149797	.1960069	-0.76	0.448	-.5434884	.2438945
20		.0800445	.2174335	0.37	0.714	-.3566836	.5167725
21		-.1103709	.0591079	-1.87	0.068	-.2290927	.008351
22		-.7439527	.2827822	-2.63	0.011	-1.311937	-.1759679
23		-2.021225	.1483314	-13.63	0.000	-2.319158	-1.723293
24		-1.474562	.5007266	-2.94	0.005	-2.4803	-.4688227
25		-1.894324	.5379483	-3.52	0.001	-2.974825	-.8138234
26		-.7679939	.3604587	-2.13	0.038	-1.491996	-.0439913
27		-.3203199	.4029102	-0.80	0.430	-1.129589	.4889491
28		-.0275888	.1456112	-0.19	0.850	-.3200575	.2648799
29		-.1669216	.2861424	-0.58	0.562	-.7416555	.4078123
30		-1.266169	.1883256	-6.72	0.000	-1.644432	-.8879054
31		-.1175657	.3082755	-0.38	0.705	-.7367553	.501624
32		-1.964808	1.044429	-1.88	0.066	-4.062606	.1329889
33		-3.590973	.8735464	-4.11	0.000	-5.345543	-1.836404
34		-2.275554	.6451338	-3.53	0.001	-3.571343	-.9797646
35		-1.181403	.3453042	-3.42	0.001	-1.874967	-.4878395
36		-1.383549	.1679959	-8.24	0.000	-1.720978	-1.046119
37		-.0884957	.1283493	-0.69	0.494	-.3462929	.1693014
38		-1.826491	.1347731	-13.55	0.000	-2.09719	-1.555791
39		-.4892242	.2468848	-1.98	0.053	-.9851069	.0066586
40		.3048617	.0728499	4.18	0.000	.1585383	.4511851
41		-1.122812	.3450629	-3.25	0.002	-1.815891	-.4297331
42		-.6310678	.3294935	-1.92	0.061	-1.292875	.0307394
44		-1.209623	.5218193	-2.32	0.025	-2.257728	-.1615186
45		-1.1109	.18893	-5.88	0.000	-1.490377	-.7314228
46		-2.028952	.701715	-2.89	0.006	-3.438388	-.6195157
47		.0592482	.1032237	0.57	0.569	-.1480827	.2665792
48		-.6054829	.327232	-1.85	0.070	-1.262748	.0517819
49		-1.266724	.206377	-6.14	0.000	-1.681244	-.8522034
50		-1.997358	.2201325	-9.07	0.000	-2.439507	-1.555209
51		-.9493606	.3246087	-2.92	0.005	-1.601356	-.2973647
53		0	(omitted)				
54		-.5213726	.1272785	-4.10	0.000	-.777019	-.2657262
55		-.4836776	.5516487	-0.88	0.385	-1.591697	.6243413
56		-2.192106	.5827897	-3.76	0.000	-3.362674	-1.021539

acc		.0028777	.0012798	2.25	0.029	.0003071	.0054484
ir		.0004439	.0004202	1.06	0.296	-.0004002	.001288
pi		-.0390221	.0662197	-0.59	0.558	-.1720282	.0939841
alcohol		.4468421	.3277729	1.36	0.179	-.211509	1.105193
crack		.0528287	.0346135	1.53	0.133	-.0166946	.1223521
poverty		-.002761	.0140407	-0.20	0.845	-.0309626	.0254406
income		.0000585	.0000443	1.32	0.193	-.0000304	.0001474
ur		-.0278237	.0364834	-0.76	0.449	-.1011027	.0454553
_cons		7.475343	1.17536	6.36	0.000	5.114563	9.836123

We can see that the difference-in-differences interaction is statistically significant from 1986 to 1992. After 1992, we fail to reject the null hypothesis.

1.4 Graph

We can save our parameter estimates using the *parmest* command, so that we can graph our difference-in-differences output.

```
parmest, label for(estimate min95 max95 %8.2f) li(parm label estimate min95 max95) saving(bf15_DD)
```

Now we need to bring the data into Stata and work with the years. First, use the Parameter estimates data frame. The data set looks like the following:

```
use ./bf15_DD.dta, replace
```

We will keep the difference-in-difference parameters, which are row 36 through row 50 observations.

```
keep in 36/50
```

Next, we will generate a year variable for each year of the parameter estimates, and then we will sort the data.

```
*Generate year
gen      year=.
replace year=1986 in 1
replace year=1987 in 2
replace year=1988 in 3
replace year=1989 in 4
replace year=1990 in 5
replace year=1991 in 6
```

parm	label	estimate	stderr	dof	t	p	min95	max95
36 1.repeal#1986.year		-0.26	.0631902	50	-4.0916297	.00015577	-0.39	-0.13
37 1.repeal#1987.year		-0.34	.1683368	50	-2.030838	.04769523	-0.68	-0.01
38 1.repeal#1988.year		-0.61	.20261371	50	-3.0133856	.00404828	-1.02	-0.20
39 1.repeal#1989.year		-0.79	.23721205	50	-3.3092765	.00173969	-1.26	-0.31
40 1.repeal#1990.year		-0.63	.17564897	50	-3.5958714	.00073893	-0.96	-0.29
41 1.repeal#1991.year		-0.55	.13935940	50	-3.9653033	.00033967	-0.83	-0.27
42 1.repeal#1992.year		-0.44	.16038747	50	-2.7584237	.00809168	-0.76	-0.12
43 1.repeal#1993.year		-0.31	.16076043	50	-1.8931946	.06645088	-0.67	0.06
44 1.repeal#1994.year		-0.12	.20667892	50	-0.5740534	.5799073	-0.53	0.30
45 1.repeal#1995.year		0.02	.22262343	50	.0944605	.92511414	-0.43	0.47
46 1.repeal#1996.year		-0.12	.20558717	50	-.5983374	.55251677	-0.64	0.29
47 1.repeal#1997.year		0.02	.25916812	50	.07993843	.93739699	-0.51	0.55
48 1.repeal#1998.year		-0.04	.34732154	50	-.10380121	.91774535	-0.73	0.66
49 1.repeal#1999.year		0.01	.35749734	50	.0411447	.96734444	-0.70	0.73
50 1.repeal#2000.year		0.04	.38734162	50	.10701357	.91520646	-0.74	0.82
51 15.fip	FIPSCODE	0.00	0	50			0.00	0.00
52 2.fip	FIPSCODE	-0.87	.29916195	50	-2.9144633	.00531766	-1.47	-0.27
53 4.fip	FIPSCODE	-0.96	.52382536	50	-1.8306455	.07311642	-2.01	0.09
54 5.fip	FIPSCODE	0.09	.06019406	50	1.3190145	.1929744	-0.04	0.22
55 6.fip	FIPSCODE	-0.30	.2236814	50	-1.3368183	.18733297	-0.74	0.15
56 8.fip	FIPSCODE	-0.82	.41811798	50	-1.9721825	.0541327	-1.66	0.02
57 9.fip	FIPSCODE	-1.68	.73500697	50	-2.2866997	.02847072	-3.16	-0.20
58 10.fip	FIPSCODE	-0.75	.48377517	50	-1.5413209	.12940336	-1.72	0.23
59 11.fip	FIPSCODE	-2.11	1.2456387	50	-1.6958055	.09614274	-4.61	0.39
60 12.fip	FIPSCODE	-1.09	.51315347	50	-2.1333727	.03782477	-2.13	-0.06
61 13.fip	FIPSCODE	-0.72	.24010818	50	-2.9319474	.00306925	-1.22	-0.23
62 15.fip	FIPSCODE	-0.79	.11042384	50	-7.1108963	4.025e-09	-1.01	-0.56
63 16.fip	FIPSCODE	-1.79	.199309	50	-8.9996907	4.827e-12	-2.19	-1.39
64 17.fip	FIPSCODE	-0.72	.45861634	50	-1.5760896	.12131193	-1.64	0.20
65 18.fip	FIPSCODE	-0.14	.19703952	50	-.70538197	.47740204	-0.52	0.23
66 19.fip	FIPSCODE	-0.15	.18502692	50	-.79424326	.44831405	-0.54	0.24
67 20.fip	FIPSCODE	0.08	.21743351	50	.3681331	.71432797	-0.36	0.52
68 21.fip	FIPSCODE	-0.11	.05910795	50	-1.8672759	.06773087	-0.23	0.01
69 22.fip	FIPSCODE	-0.74	.28782118	50	-2.5305259	.01120663	-1.31	-0.18
70 23.fip	FIPSCODE	-2.02	.14833141	50	-13.626415	1.845e-18	-2.32	-1.72
71 24.fip	FIPSCODE	-1.47	.50072655	50	-2.9448439	.004893	-2.48	-0.47
72 25.fip	FIPSCODE	-1.89	.53794625	50	-3.5213879	.00039545	-2.97	-0.81
73 26.fip	FIPSCODE	-0.77	.36348959	50	-2.1306718	.03809399	-1.49	-0.04
74 27.fip	FIPSCODE	-0.32	.40291021	50	-.79501555	.43062227	-1.13	0.49
75 28.fip	FIPSCODE	-0.03	.1458112	50	-.18948917	.85049217	-0.32	0.26
76 29.fip	FIPSCODE	-0.17	.29614238	50	-.58335162	.56227893	-0.74	0.41

Figure 1.1: Parameter Estimates Output

```

replace year=1992 in 7
replace year=1993 in 8
replace year=1994 in 9
replace year=1995 in 10
replace year=1996 in 11
replace year=1997 in 12
replace year=1998 in 13
replace year=1999 in 14
replace year=2000 in 15

sort year

```

Now we will graph our parameters using a twoway graph

```

twoway (scatter estimate year, mlabel(year) mlabsize(vsmall) msize(tiny)) ///
      (rcap min95 max95 year, msize(vsmall)), ytitle(Repeal x year estimated coefficient) ///
      yscale(titlegap(2)) yline(0, lwidth(thin) lcolor(black)) xtitle(Year) ///
      xline(1986 1987 1988 1989 1990 1991 1992, lwidth(vvvthick)) ///
      lpattern(solid) lcolor(ltblue)) xscale(titlegap(2)) ///
      title(Estimated effect of abortion legalization on gonorrhea) ///
      subtitle(Black females 15-19 year-olds) ///
      note(W/o XI; Whisker plots are estimated coefficients of DD estimator from Column b) ///
      legend(off)

```

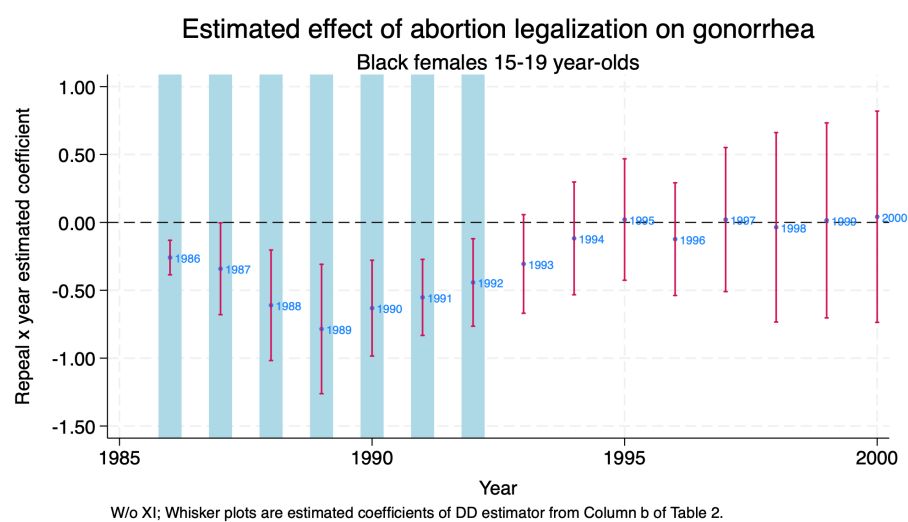


Figure 1.2: Difference-in-Differences Parameter Estimates 1986-2000

Chapter 2

Placebo Tests: False Treatment

We have two major types of difference-in-differences placebo tests:

1. False Treatment
2. False Timing

2.1 False Treatment

Our repeal states are Alaska (02), California (06), Hawaii (15), New York (36), Washington (53), so let's do false treatment placebo. We will assign Florida (12), Minnesota (27), Nevada (32), Ohio (39), and Pennsylvania (42) as our false treatment/repeal states.

```
use "/Users/Sam/Desktop/Econ 672/Data/abortion.dta", clear

gen repeal_fake = 0
replace repeal_fake = 1 if fip == 12
replace repeal_fake = 1 if fip == 27
replace repeal_fake = 1 if fip == 32
replace repeal_fake = 1 if fip == 39
replace repeal_fake = 1 if fip == 42

tab fip repeal_fake

reg lnr i.repeal_fake##i.year i.fip acc ir pi alcohol crack poverty income ur if bf15==1 [aweight
```

(384 real changes made)

(384 real changes made)

(384 real changes made)

(384 real changes made)

(384 real changes made)

FIPSCODE	repeal_fake		Total
	0	1	
1	384	0	384
2	384	0	384
4	384	0	384
5	384	0	384
6	384	0	384
8	384	0	384
9	384	0	384
10	384	0	384
11	384	0	384
12	0	384	384
13	384	0	384
15	384	0	384
16	384	0	384
17	384	0	384
18	384	0	384
19	384	0	384
20	384	0	384
21	384	0	384
22	384	0	384
23	384	0	384
24	384	0	384
25	384	0	384
26	384	0	384
27	0	384	384
28	384	0	384
29	384	0	384
30	384	0	384
31	384	0	384
32	0	384	384
33	384	0	384
34	384	0	384
35	384	0	384
36	384	0	384

37		384		0		384
38		384		0		384
39		0		384		384
40		384		0		384
41		384		0		384
42		0		384		384
44		384		0		384
45		384		0		384
46		384		0		384
47		384		0		384
48		384		0		384
49		384		0		384
50		384		0		384
51		384		0		384
53		384		0		384
54		384		0		384
55		384		0		384
56		384		0		384
-----+-----+-----						
Total		17,664		1,920		19,584

(sum of wgt is 43,100,087)

note: 42.fip omitted because of collinearity.

Linear regression

Number of obs	=	736
F(26, 50)	=	.
Prob > F	=	.
R-squared	=	0.8430
Root MSE	=	.25494

(Std. err. adjusted for 51 clusters in fip)

-----+-----+-----						
		Robust				
lnr		Coefficient	std. err.	t	P> t	[95% conf. interval]
-----+-----+-----						
1.repeal_fake		-.2619235	.3063582	-0.85	0.397	-.877262 .353415
year						
1986		-.0473925	.0623881	-0.76	0.451	-.1727026 .0779176
1987		-.282811	.1071218	-2.64	0.011	-.4979714 -.0676506
1988		-.3047447	.1380771	-2.21	0.032	-.5820807 -.0274087
1989		-.3381935	.1805399	-1.87	0.067	-.7008186 .0244316
1990		-.3411266	.2247542	-1.52	0.135	-.7925586 .1103054
1991		-.3638668	.2180394	-1.67	0.101	-.8018117 .0740781
1992		-.5641131	.2553435	-2.21	0.032	-1.076986 -.0512405
1993		-.6982166	.25845	-2.70	0.009	-1.217329 -.1791045

1994		-.704213	.2728245	-2.58	0.013	-1.252197	-.1562289
1995		-.938756	.3081557	-3.05	0.004	-1.557705	-.3198071
1996		-1.085633	.3629182	-2.99	0.004	-1.814576	-.3566907
1997		-1.069773	.4168856	-2.57	0.013	-1.907112	-.2324331
1998		-1.006071	.4858285	-2.07	0.044	-1.981887	-.0302559
1999		-1.017142	.5311595	-1.91	0.061	-2.084007	.049723
2000		-1.062719	.5999853	-1.77	0.083	-2.267825	.1423874
repeal_fake#							
year							
1 1986		.1961008	.1580668	1.24	0.221	-.1213857	.5135873
1 1987		.3019751	.1659506	1.82	0.075	-.0313465	.6352967
1 1988		.1925532	.1376346	1.40	0.168	-.083894	.4690004
1 1989		.2587087	.163867	1.58	0.121	-.070428	.5878453
1 1990		.1725233	.1545046	1.12	0.269	-.1378082	.4828549
1 1991		.0588022	.1357068	0.43	0.667	-.2137729	.3313773
1 1992		-.0298446	.148425	-0.20	0.841	-.3279649	.2682758
1 1993		-.0782904	.1372765	-0.57	0.571	-.3540183	.1974375
1 1994		-.0599632	.123157	-0.49	0.628	-.3073313	.1874049
1 1995		.0068644	.145227	0.05	0.962	-.2848327	.2985614
1 1996		.006097	.1133758	0.05	0.957	-.221625	.233819
1 1997		-.0625821	.1155086	-0.54	0.590	-.294588	.1694237
1 1998		-.0846898	.1382085	-0.61	0.543	-.3622897	.1929101
1 1999		-.0037381	.1298435	-0.03	0.977	-.2645364	.2570603
1 2000		.0295735	.1186556	0.25	0.804	-.2087532	.2679002
fip							
2		-1.67931	.7013951	-2.39	0.020	-3.088104	-.2705167
4		-.4782904	.4415301	-1.08	0.284	-1.36513	.4085488
5		.0570652	.0799359	0.71	0.479	-.1034908	.2176211
6		-.9959097	.5272526	-1.89	0.065	-2.054928	.0631083
8		-.5320145	.4589932	-1.16	0.252	-1.45393	.3899005
9		-.9013381	.7003092	-1.29	0.204	-2.307951	.5052743
10		-.3408189	.5989758	-0.57	0.572	-1.543897	.8622595
11		-1.319298	1.324266	-1.00	0.324	-3.979165	1.340569
12		-.4700227	.2584934	-1.82	0.075	-.989222	.0491766
13		-.4658355	.2632261	-1.77	0.083	-.9945406	.0628696
15		-1.712506	.4372031	-3.92	0.000	-2.590654	-.8343573
16		-1.698102	.1825109	-9.30	0.000	-2.064686	-1.331518
17		-.3702913	.4493088	-0.82	0.414	-1.272755	.532172
18		.1378953	.1998675	0.69	0.493	-.2635504	.5393411
19		.0936967	.1653353	0.57	0.573	-.238389	.4257823
20		.3886963	.1730131	2.25	0.029	.0411892	.7362035
21		-.0840023	.056987	-1.47	0.147	-.1984641	.0304594
22		-.623181	.2715091	-2.30	0.026	-1.168523	-.0778389
23		-2.043909	.2082448	-9.81	0.000	-2.462181	-1.625637

24		-.9304013	.5056388	-1.84	0.072	-1.946007	.0852041
25		-1.362687	.6297522	-2.16	0.035	-2.627582	-.0977927
26		-.4230176	.3390352	-1.25	0.218	-1.10399	.2579546
27		.2942861	.1721111	1.71	0.093	-.0514092	.6399814
28		-.1631074	.132844	-1.23	0.225	-.4299325	.1037176
29		.1893571	.2860834	0.66	0.511	-.3852584	.7639725
30		-1.350451	.248263	-5.44	0.000	-1.849102	-.8518006
31		.2705925	.2813189	0.96	0.341	-.294453	.8356381
32		-1.022858	.8797807	-1.16	0.250	-2.789949	.744234
33		-3.093851	1.071677	-2.89	0.006	-5.246377	-.9413251
34		-1.651775	.6231486	-2.65	0.011	-2.903406	-.4001445
35		-1.024749	.2929583	-3.50	0.001	-1.613173	-.4363253
36		-2.118819	.4655078	-4.55	0.000	-3.053819	-1.18382
37		.0543435	.1264402	0.43	0.669	-.1996191	.308306
38		-1.83464	.2001633	-9.17	0.000	-2.23668	-1.4326
39		.0414655	.0717687	0.58	0.566	-.1026862	.1856172
40		.3004358	.0858056	3.50	0.001	.1280901	.4727815
41		-.7933867	.3571316	-2.22	0.031	-1.510707	-.0760667
42		0	(omitted)				
44		-.5405153	.5104011	-1.06	0.295	-1.565686	.4846555
45		-.9602952	.2294899	-4.18	0.000	-1.421239	-.4993512
46		-1.127867	.4597007	-2.45	0.018	-2.051204	-.2045314
47		.1312519	.0886643	1.48	0.145	-.0468356	.3093394
48		-.3837915	.3086508	-1.24	0.220	-1.003735	.2361518
49		-.966726	.2534117	-3.81	0.000	-1.475718	-.4577335
50		-1.942943	.3250246	-5.98	0.000	-2.595774	-1.290112
51		-.5903806	.3069945	-1.92	0.060	-1.206997	.026236
53		-.8368507	.4010224	-2.09	0.042	-1.642328	-.0313735
54		-.5250537	.1220392	-4.30	0.000	-.7701766	-.2799308
55		.1296001	.5587953	0.23	0.818	-.9927732	1.251973
56		-1.497547	.4613406	-3.25	0.002	-2.424176	-.5709167
acc		.0017173	.0010087	1.70	0.095	-.0003087	.0037433
ir		-.0001316	.0002427	-0.54	0.590	-.0006192	.000356
pi		-.1012491	.0940607	-1.08	0.287	-.2901756	.0876774
alcohol		.3243983	.3896246	0.83	0.409	-.4581858	1.106982
crack		.0406948	.0415672	0.98	0.332	-.0427954	.124185
poverty		.0089868	.0164328	0.55	0.587	-.0240195	.0419931
income		.0000331	.0000418	0.79	0.432	-.0000508	.0001171
ur		-.0143629	.0374348	-0.38	0.703	-.0895528	.0608271
_cons		7.790753	1.229231	6.34	0.000	5.32177	10.25974

Again, we will save the parameter estimates, keep the difference-in-difference parameters, and graph our results.

```

parwest, label for(estimate min95 max95 %8.2f) li(parm label estimate min95 max95) ///
    saving(bf15_DD_false.dta, replace)

use ./bf15_DD_false.dta, replace

```

Keep the Diff-in-Diff Interaction Parameter Estimates and generate a year variable

```

keep in 36/50

gen      year=.
replace year=1986 in 1
replace year=1987 in 2
replace year=1988 in 3
replace year=1989 in 4
replace year=1990 in 5
replace year=1991 in 6
replace year=1992 in 7
replace year=1993 in 8
replace year=1994 in 9
replace year=1995 in 10
replace year=1996 in 11
replace year=1997 in 12
replace year=1998 in 13
replace year=1999 in 14
replace year=2000 in 15

```

Finally, we will sort and graph our data.

```

sort year

*Graph Parameter Estimates with Confidence Intervals
twoway (scatter estimate year, mlabel(year) mlabsize(vsmall) msize(tiny)) ///
    (rcap min95 max95 year, msize(vsmall)), ytitle(Repeal x year estimated coefficient) ///
    yscale(titlegap(2)) yline(0, lwidth(thin) lcolor(black)) xtitle(Year) ///
    xline(1986 1987 1988 1989 1990 1991 1992, lwidth(vvvthick) ///
    lpattern(solid) lcolor(ltblue)) xscale(titlegap(2)) ///
    title(Placebo Test: Fake Treatment of abortion legalization on gonorrhea) ///
    subtitle(Black females 15-19 year-olds) ///
    note(W/o XI; Wh

```

In support of Cunningham and Cornwell's hypothesis, we fail to reject the null hypothesis at the 5 percent level across the years.

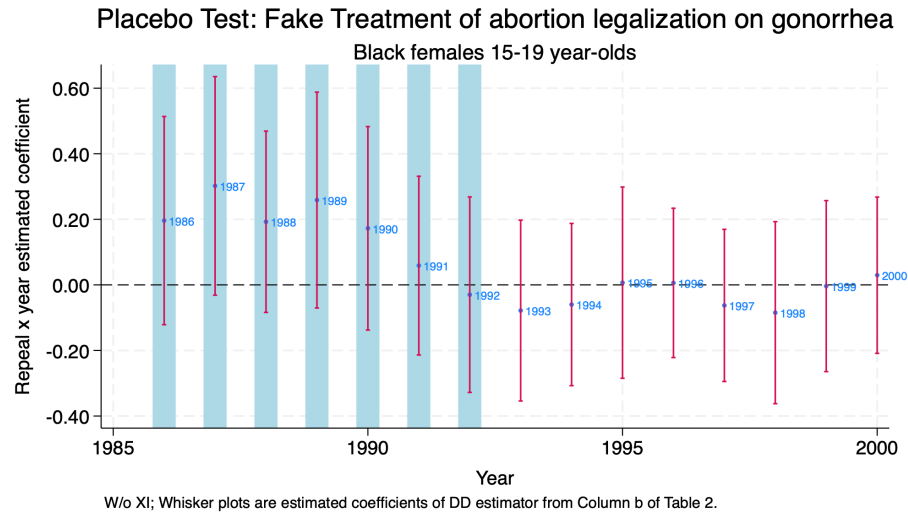


Figure 2.1: Placebo Test: False Treatment

Chapter 3

*xt**did**regress* commands

In addition to using *xtreg* or *reg* for estimating our δ_{DD}^{AET} , we can use some built in Stata commands instead. These commands include *xtdidregress* and *didregress*.

For additional information we can use the *help* command.

```
help didregress
```

Stata's help page provides us with the following information:

For repeated cross-sections:

```
didregress (ovar omvarlist) (tvar[, continuous]) [if] [in] [weight],  
group(groupvars) [time(timevar) options]
```

For panel data:

```
xtdidregress (ovar omvarlist) (tvar[, continuous]) [if] [in] [weight],  
group(groupvars) [time(timevar) options]
```

Where 1. *ovar* is our outcome of interest 2. *omvarlist* is a list of our explanatory covariates 3. *tvar* is a binary indicating treatment status or a continuous indicating intensity of treatment 4. *groupvars* are categorical variables indicating treatment level, such as individual, state, etc. 5. *timevar* is our time variable, such as year

3.1 World Development Index

We will use the World Development Index to assess the impact of a trade policy in 2000 on total amount of trade (imports + exports) in millions of 2005 US dollars. These data are provided by Princeton.

```
cd "/Users/Sam/Desktop/Econ 672/Data/"
use wdipol, clear
describe
```

```
/Users/Sam/Desktop/Econ 672/Data
```

```
Contains data from wdipol.dta
```

```
Observations:      4,542
```

```
Variables:         12
```

```
26 Mar 2026 19:15
```

Variable name	Storage type	Display format	Value label	Variable label
year	int	%10.0g		Year
country	str24	%24s		Country Name
gdppc	double	%10.0g		GDP per capita, PPP (constant 2005 international \$)
unempf	double	%10.0g		Unemployment, female (% of female labor force)
unempm	double	%10.0g		Unemployment, male (% of male labor force)
unemp	double	%10.0g		Unemployment, total (% of total labor force)
export	double	%10.0g		Exports of goods and services (constant 2005 US\$)
import	double	%10.0g		Imports of goods and services (constant 2005 US\$)
polity	byte	%8.0g		polity (original)
polity2	byte	%8.0g		polity2 (adjusted)
trade	float	%9.0g		Imports + Exports
id	float	%9.0g		group(country)

```
Sorted by:
```

We will focus on years 1990 forward.

```
drop if year < 1990
sum export import trade, detail
```

```
(1,136 observations deleted)
```

```
Exports of goods and services (constant 2005 US$)
```

	Percentiles	Smallest		
1%	91.64278	45.02022		
5%	245.0359	48.62297		
10%	543.8475	52.79048	Obs	2,814
25%	2094.341	53.24336	Sum of wgt.	2,814
50%	8836.828		Mean	75432.49
		Largest	Std. dev.	182677.6
75%	57261.36	1649300		
90%	198699.2	1665600	Variance	3.34e+10
95%	399759.2	1677840	Skewness	4.764539
99%	1036798	1776900	Kurtosis	31.56892

Imports of goods and services (constant 2005 US\$)

	Percentiles	Smallest		
1%	264.6554	94.23958		
5%	538.7271	110.3702		
10%	1031.108	129.2793	Obs	2,814
25%	2514.115	135.4336	Sum of wgt.	2,814
50%	10218.4		Mean	74643.16
		Largest	Std. dev.	194233
75%	56131.35	2144000		
90%	182048.5	2151500	Variance	3.77e+10
95%	371224.2	2184900	Skewness	5.811369
99%	943900	2203200	Kurtosis	47.77702

Imports + Exports

	Percentiles	Smallest		
1%	414.4894	139.2598		
5%	828.4978	166.8271		
10%	1613.88	183.3966	Obs	2,814
25%	4867.792	193.1237	Sum of wgt.	2,814
50%	19421.02		Mean	150075.7
		Largest	Std. dev.	374324.7
75%	115498.2	3750800		
90%	382896.8	3757600	Variance	1.40e+11
95%	759593.4	3793300	Skewness	5.158498
99%	2001955	3961800	Kurtosis	37.13028

Looking at combined imports and exports, the median amount of trade is 19.4 billion 2005 US dollars.

Now, we set our post period at 2000 or later. An indicator for when trade relations were normalized between the U.S. and China and China's ascension into the World Trade Organization (WTO).

```
gen post = .
replace post = 0 if year < 2000
replace post = 1 if year >= 2000
```

Next we will assign the treatment countries, which are the following western countries: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, and United States.

```
gen treatment = 0
replace treatment = 1 if country == "Australia" | country == "Austria" | country == "B
```

Next, we will set up the difference-in-differences interaction.

```
gen treat_post=treatment*post
```

Now we need to establish our panel, but we need to use the *encode* command on countries so we can use the *xtset* command.

```
*Encode country
encode country, gen(countrynum)

*Set Panel
sort countrynum year
xtset countrynum year
```

```
Panel variable: countrynum (unbalanced)
Time variable: year, 1990 to 2012, but with a gap
Delta: 1 unit
```

Our panel is unbalanced, which means we do have gaps among some of the countries.

We will first start with our tried and true method with *xtreg* and use the *fe* option, and we will cluster our standard errors at the unit of observation level.

```
xtreg trade treat_post i.year, fe vce (cluster countrynum)
```

```

Fixed-effects (within) regression      Number of obs   =    2,814
Group variable: countrysum            Number of groups =    153

R-squared:                             Obs per group:
    Within = 0.2476                      min =          1
    Between = 0.2331                     avg =         18.4
    Overall = 0.1877                     max =          23

                                         F(23, 152)      =    2.88
corr(u_i, Xb) = 0.2187                 Prob > F         =    0.0001

```

(Std. err. adjusted for 153 clusters in countrysum)

		Robust				
	trade	Coefficient	std. err.	t	P> t	[95% conf. interval]
treat_post		254619.5	84665.83	3.01	0.003	87345.69 421893.2
year						
1991		11417.79	3273.274	3.49	0.001	4950.798 17884.77
1992		14691.17	4183.539	3.51	0.001	6425.782 22956.57
1993		17823.51	5089.347	3.50	0.001	7768.513 27878.5
1994		27212.01	6768.259	4.02	0.000	13840.01 40584.02
1995		36908.65	8577.682	4.30	0.000	19961.77 53855.52
1996		43697.3	9997.622	4.37	0.000	23945.06 63449.54
1997		53882.39	12313.33	4.38	0.000	29555.02 78209.75
1998		60352.96	13818.67	4.37	0.000	33051.5 87654.42
1999		67276.08	15777.22	4.26	0.000	36105.13 98447.04
2000		43940.78	12961.89	3.39	0.001	18332.06 69549.5
2001		45574.41	13023.31	3.50	0.001	19844.34 71304.49
2002		50357.95	14195.9	3.55	0.001	22311.21 78404.7
2003		58200.38	16052.8	3.63	0.000	26484.97 89915.79
2004		74493.25	18905.12	3.94	0.000	37142.52 111844
2005		87670.9	21325.73	4.11	0.000	45537.78 129804
2006		104994.9	24623.44	4.26	0.000	56346.51 153643.3
2007		121106.3	28005.54	4.32	0.000	65775.88 176436.6
2008		128370.5	29721.71	4.32	0.000	69649.48 187091.5
2009		104644.4	27073.14	3.87	0.000	51156.19 158132.7
2010		131099.8	33071.58	3.96	0.000	65760.48 196439.1
2011		149272.6	37479.36	3.98	0.000	75224.91 223320.4
2012		129456.8	24816.73	5.22	0.000	80426.54 178487.1
_cons		56946.96	18846.82	3.02	0.003	19711.41 94182.51
sigma_u		295226.32				
sigma_e		132684.6				

```
rho | .83195335 (fraction of variance due to u_i)
```

Our difference-in-difference estimator indicates that the change in trade policy increased total trade by 255 billion 2005 US dollars, which is statistically significant at the 1% level.

To visualize our parallel trends, we need to use *egen* to get the mean outcome by year for treatment and comparison groups. Next, we will visually inspect the parallel trends assumption with *twoway line* charts.

```
bysort year treatment: egen mean_trade = mean(trade)

twoway line mean_trade year if treatment == 0, sort || ///
      line mean_trade year if treatment == 1, sort lpattern (dash) ///
      legend(label(1 "Control") label(2 "Treated")) ///
      xline(2000) title("Check Parallel Trends Assumption")
```

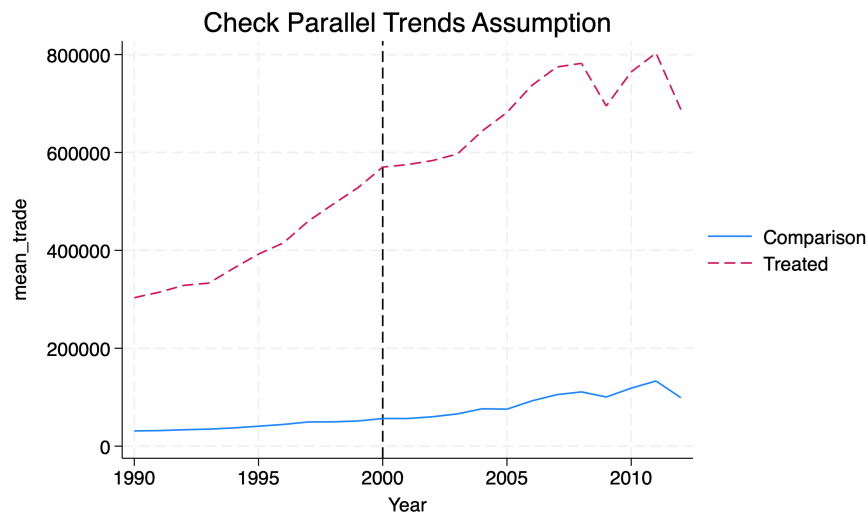


Figure 3.1: Inspect the Parallel Trends Assumption

3.2 *xtdidregress*

We have other commands for estimating difference-in-differences, such as the *xt-didregress* and *didregress* commands. These commands will estimate our δ_{DD}^{ATE} of interest.

Why should we use this command? They provide helpful post-estimation commands for testing the parallel trends assumption!

```
xtdidregress (trade) (treat_post), group(countrynum) time(year)
```

Treatment and time information

Time variable: year

Control: treat_post = 0

Treatment: treat_post = 1

	Control	Treatment
Group		
countrynum	132	21
Time		
Minimum	1990	2000
Maximum	2006	2000

Difference-in-differences regression
Data type: Longitudinal

Number of obs = 2,814

(Std. err. adjusted for 153 clusters in countrynum)

	trade	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
ATET						
treat_post						
(1 vs 0)		254619.5	84665.83	3.01	0.003	87345.69 421893.2

Note: ATET estimate adjusted for panel effects and time effects.

Note: Treatment occurs at different times.

We get the same result, such that trade increases \$254,619.5 millions or 255 billion 2005 US dollars

We can also add some covariates in case the parallel trends is conditional on controlling for covariates.

```
xtdidregress (trade unemp polity) (treat_post), group(countrynum) time(year)
```

Treatment and time information

Time variable: year
 Control: treat_post = 0
 Treatment: treat_post = 1

	Control	Treatment
Group		
countrynum	132	21
Time		
Minimum	1990	2000
Maximum	2006	2000

Difference-in-differences regression
 Data type: Longitudinal

Number of obs = 2,804

(Std. err. adjusted for 153 clusters in countrynum)

		Robust				
trade	Coefficient	std. err.	t	P> t	[95% conf. interval]	
ATET						
treat_post						
(1 vs 0)	253104.1	85410.71	2.96	0.004	84358.69	421849.5

Note: ATET estimate adjusted for covariates, panel effects, and time effects.
 Note: Treatment occurs at different times.

Our estimate is little change after controlling for unemployment and political system of the country. Our estimate is $\hat{\delta}_{DD}^{ATET} = \$253B$.

Chapter 4

Postestimation Command

The question remains if our parallel trends assumption holds. We visually inspected the parallel trends assumption with a graph, but we did not test any null hypotheses. With the *xtdidregress* and *didregress* commands we have two helpful commands: *estat ptrends* and *estat trendplots*.

4.1 *estat ptrends*

Let's test the null hypothesis that the trends between treatment and comparison are the same. We will use the diff-in-diff without covariates first

```
quietly xtdidregress (trade) (treat_post), group(countrynum) time(year)
estat ptrends
```

Parallel-trends test (pretreatment time period)
H0: Linear trends are parallel

```
F(1, 152) = 9.09
Prob > F = 0.0030
```

Our F -statistic is 9.09 and we reject the null hypothesis at the 1% level. Now let's add the covariates to see if parallel trends are conditional on additional covariates of unemployment and political system.

```
quietly xtdidregress (trade unemp polity) (treat_post), group(countrynum) time(year)
estat ptrends
```

Parallel-trends test (pretreatment time period)
H0: Linear trends are parallel

$F(1, 152) = 8.85$
Prob > F = 0.0034

Our F -statistic barely changes and we reject the null hypothesis at the 1% level.

4.2 *estat trendsplot*

Even though we reject the null hypothesis at the 1% level, we can still provide a visualization that is easier to implement than our *twoway line* graph above. With *estat trendsplot*, the syntax is more concise and less likely to have an operator error. With the *twoway line* command, we need to make sure to graph both treatment and comparison lines.

We will compare the parallel trends with and without covariates. The command is the same regardless if we have covariates or not.

```
quietly xtdidregress (trade) (treat_post), group(countrynum) time(year)
estat trendsplots, ytitle(Trade in millions of 2005 USD) xlabel(1990(2)2012)
```

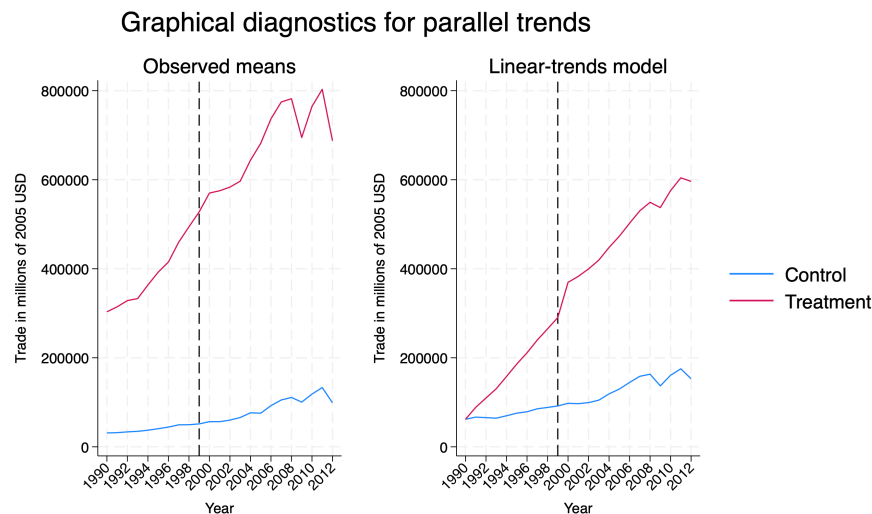


Figure 4.1: Parallel trends without covariates

We can clearly see that the parallel trend assumption does not hold. This visually confirms what we saw with our F -test.

Next, we will add covariates.

```
quietly xtdidregress (trade) (treat_post), group(countrynum) time(year)
estat trendplots, ytitle(Trade in millions of 2005 USD) xlabel(1990(2)2012)
```

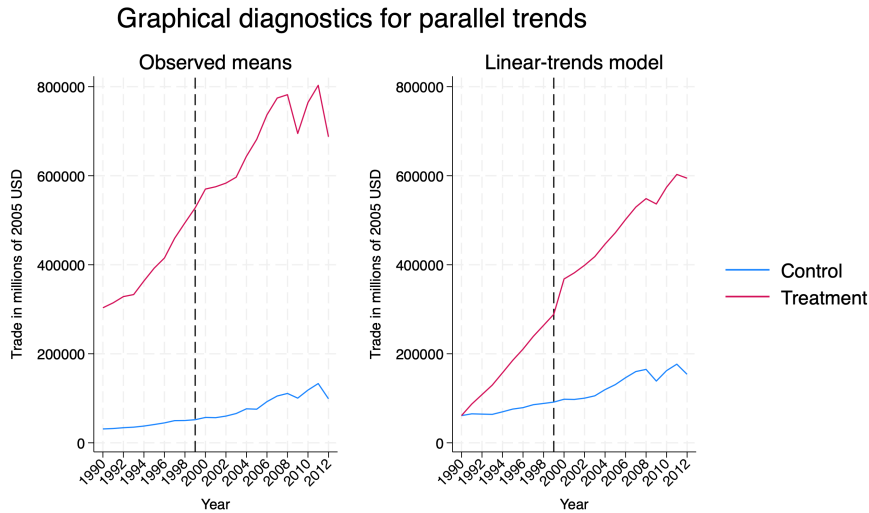


Figure 4.2: Parallel trends with covariates

Again, we clearly see that the parallel trends assumption does not hold even if we add covariates.

Chapter 5

Placebo Tests: False Timing

Our second major type of placebo test for difference-in-differences estimator is false timing. We can accomplish this by changing our t to $t - n$.

5.1 False Timing

For our trade example, our treatment date is the year 2000, but let's set the date to 1995.

```
gen post2 = .  
replace post2 = 0 if year < 1995  
replace post2 = 1 if year >= 1995  
  
gen treat_post2=treatment*post2
```

We should expect no statistically significant difference between treatment and comparison for our placebo test.

```
xtddidregress (trade) (treat_post2), group(countrynum) time(year)
```

Treatment and time information

```
Time variable: year  
Control:      treat_post2 = 0  
Treatment:    treat_post2 = 1  
-----  
                | Control Treatment  
-----+-----
```

Group			
countrynum		132	21

Time			
Minimum		1990	1995
Maximum		2006	2000

Difference-in-differences regression
Data type: Longitudinal

Number of obs = 2,814

(Std. err. adjusted for 153 clusters in countrynum)

			Robust				
trade		Coefficient	std. err.	t	P> t	[95% conf. interval]	

ATET							
treat_post2							
(1 vs 0)		253834	83426.64	3.04	0.003	89008.47	418659.5

Note: ATET estimate adjusted for panel effects and time effects.

Note: Treatment occurs at different times.

We reject the null hypothesis that the parameter is equal to zero, so our placebo test fails. There is something going on with these countries that may increase trade relative to other countries besides the trade policy change.

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